

Experiment Results Summary

Language Modelling Is Compression | Llama 2 7B on enwik9 | Updated 31 August 2026

This document summarises the completed experiments and identifies the supporting files retained in the repository. It distinguishes model-based coding-length estimates from archive sizes measured from decodable files.

1. Full 1 GB Llama 2 evaluation

Item	Recorded result
Model and input	meta-llama/Llama-2-7b-hf; complete 1,000,000,000-byte enwik9 file
Processing	488,282 independent 2,048-byte chunks; 338,899,783 scored tokens
Model-only result	BPC 0.7073548; theoretical compression rate 8.8419%
Paper-comparable result	8.8967% after adding one bit for each byte changed by ASCII mapping
Other checks	Top-100 hit rate 96.9410%; 0 skipped chunks; 0 truncated chunks
Runtime	38,170.50 seconds (approximately 10.6 hours)

The 8.8967% result is a model-based estimate used for comparison with the reference study. No 1 GB range-coded archive was generated, and model parameter storage is excluded.

2. Conventional compression baselines

Method	Archive size	Rate	Verification
gzip level 9	322,591,988 bytes	32.2592%	Round trip verified
bzip2 level 9	253,977,891 bytes	25.3978%	Round trip verified
LZMA2/XZ preset 9	213,370,900 bytes	21.3371%	Round trip verified

The baselines operated on the original, unchunked enwik9 byte stream. Each retained archive was decompressed and the reconstructed SHA-256 matched the source file. Original compression timings were not retained.

3. Practical range-coding validation

Item	Recorded result
Input	First 1,024,000 bytes of enwik9; 500 independent 2,048-byte chunks
Method	Llama 2 7B with constriction 0.4.2; Top-100 range-coded categories plus an escape-token side stream
Model-only estimate	8.8705%
Actual archive	107,884 bytes; actual compression rate 10.5355%
Archive components	76,500-byte range stream; 22,058-byte escape stream; 6,608-byte header and framing metadata; 2,718-byte high-bit mask
Lossless check	Verified: reconstructed SHA-256 was identical to input SHA-256
Runtime	Encode 7,452.81 seconds; decode 7,392.19 seconds; total approximately 4.12 hours

The practical experiment produced a complete decodable archive at the tested scale. A full 1 GB range-coding round trip was not run; a linear extrapolation from the completed subset gives roughly 4,000 hours for the current sequential implementation.

4. Repository evidence

Location	Contents
code/	Three Python scripts used for model scoring, conventional baselines and practical range coding
slurm/	Five CSF submission scripts for smoke, full and baseline jobs
results/	Five JSON records covering the full evaluation, smoke tests, conventional baselines and practical coding
logs/	Raw SLURM output records for four successful model-scoring and range-coding jobs
environment/	Python, PyTorch, Transformers, CUDA and Conda package records
checksums/	SHA-256 records for the enwik9 input and experiment scripts
notebooks/ and figures/	Source notebook and exported workflow figure

The enwik9 input, Llama 2 model weights, large conventional archives, intermediate checkpoints and custom .Imac archives are not distributed. Their relevant sizes, checksums and verification outcomes are retained in the repository records.

Primary records: results/llama2_7b_enwik9_full.json, results/enwik9_classical_baselines_reconstructed.json and results/llama2_enwik9_1mb.json.